

BRANCH DOCUMENT — IRIS-3

Branch: branch_20260512_151747 | Base: main-v1.0

Generated: 2026-05-12 15:17

Decision: GO_WITH_CONSTRAINTS | Confidence: 78%

CHANGE REQUEST

modifichiamo il materiale della struttura facendola in marmo mantenendo le dimensioni

CASCADE ISSUES IDENTIFIED

[CRITICAL] ISSUE-001 — Mass budget violation due to marble material

Subsystem: structure | Req: L1-MAS-001 | Type: constraint_violation

Marble is significantly heavier than the original structural material, causing mass budget overruns and violating L1-MAS-001 requirement.

[MAJOR] ISSUE-002 — Shock test required due to material change

Subsystem: mechanical | Req: L1-MEC-003 | Type: missing_test

The use of marble as a structural material necessitates retesting for shock response, which is currently pending (NCR-2024-041).

[MINOR] ISSUE-003 — Thermal conductivity impact on ΔT

Subsystem: thermal | Req: None | Type: reanalysis_required

Marble has different thermal properties compared to the original material, potentially affecting thermal gradients and violating L1-THM-001 requirement.

MISSING ECSS TESTS — NOW REQUIRED

ECSS-E-ST-32C Rev.1 §9 — Shock test per launcher SRS ICD

Reason: Material change requires retesting for shock response.

SUBSYSTEM STATUS — UPDATED

Subsystem	Score	Status	Detail	Changed
optical	92%	OK	MTF 0.42 > threshold 0.40	—
thermal	61%	WARN	ΔT 14.8°C — margin < 5%	—
mechanical	88%	OK	1st mode 112 Hz > 80 Hz req	—
structure	91%	OK	Mass 33.8 kg / budget 35 kg	—
solar	89%	OK	Power margin +18%	—
startracker	67%	WARN	Stray light analysis pending	—
dataproc	71%	WARN	Compression ratio TBC	—

AFFECTED REQUIREMENTS

- L1-MAS-001 — verification evidence invalidated, re-analysis required
- L1-THM-001 — verification evidence invalidated, re-analysis required